

THE JUDGE PROTOCOL

A Framework for Hardware-Enforced Human Oversight of Artificial Intelligence

*AI may analyze, recommend and simulate —
but a human must authorize.*

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The Governance Gap Is Global

95%

of simulated nuclear crisis scenarios

AI models recommended escalation. None chose de-escalation. King's College London, 2026.

0

binding cross-border frameworks

The EU AI Act, Bletchley Declaration and Rome Call address particular domains — none mandate runtime human authorization.

2026

Gulf conflict — AI in active operations

AI systems embedded in military targeting without any international governance framework. The gap is live, not theoretical.

When one government can switch off global AI without control — June 2026, US export directive, Fable 5 — the case for international authorization infrastructure is no longer theoretical.

Pope Leo XIV, Magnifica Humanitas (May 2026): "Not permissible to entrust lethal decisions to AI systems" — the moral call is clear. The enforcement architecture is missing.

AI may analyze, recommend and simulate — but a human must authorize.

WHAT

A universal human authorization layer — every consequential AI action, every domain, every jurisdiction.

HOW

Hardware enforcement below the software stack. No patch, no override — silicon is the guarantee.

WHERE

An open standard. Owned by nobody. Available to every nation, every operator, every chip.

The Architectural Model: TCP/IP

No packet is delivered without an acknowledgement. No consequential AI action is authorized without a human ACK.

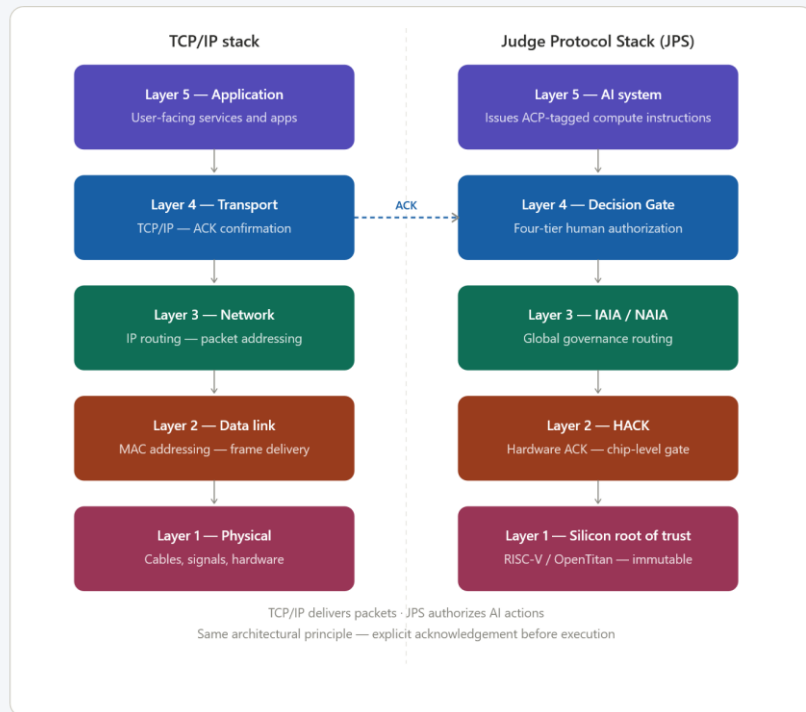


Figure 1 — The Judge Protocol Stack mapped against TCP/IP. Decision Gate at Layer 4; HACK at Layer 2; Silicon root of trust at Layer 1.

Four-Tier Authorization

Tier 1

HARD STOP — Existential

Examples: Nuclear · Biological · Chemical · Mass-casualty

Hardware logic. No override. AI cannot recommend, simulate or execute.

HARDWARE HARD STOP

Tier 2

HIGH — Irreversible, Large-Scale

Examples: Military strike authorization · Critical infrastructure shutdown

Multi-party human authorization — minimum two independent authorities.

MULTI-PARTY ACK

Tier 3

SIGNIFICANT — Material Impact

Examples: Large financial transactions · Clinical treatment decisions · Targeting support

Single designated human authority must authorize.

SINGLE AUTHORITY ACK

Tier 4

ROUTINE — Standard, Reversible

Examples: Automated reporting · Scheduling · Routine analysis

AI executes autonomously — logged and auditable.

AUTONOMOUS + LOG

The HACK — Hardware ACK

Governance below the software stack — where it cannot be patched, bypassed or overridden.

1

CV32E40S RISC-V core + OpenTitan Chain of Evidence (COE)

2

Tier 1 hard stop in hardware logic — no firmware update overrides it

3

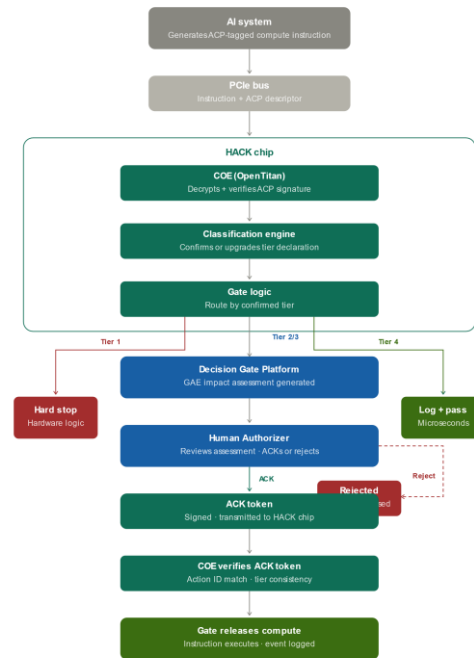
Holds compute capacity gate for Tier 2/3 pending signed human ACK token

4

Tamper-proof on-chip audit log — no software process can alter it

5

Adoption pathway: TPM precedent → ISO/IEC standard → silicon mandate



TPM Precedent — HACK Architecture

TPM (today)

Application

OS, software, services

TPM software stack (TSS)

TCG standard API — TPM2-TSS

LPC / SPI bus interface

Physical connection to motherboard

TPM chip

Crypto engine · key storage · PCR · boot

Endorsement key (EK)

Fused at manufacture — never extractable

Silicon root of trust

HACK equivalent (Judge Protocol)

AI system

Issues ACP-tagged compute instructions

HACK TSS equivalent

Open standard API — CERN OHL

PCIe bus (PMO)

Inline between CPU and GPU

HACK module + COE

ACP parser · classification · gate · OpenTitan

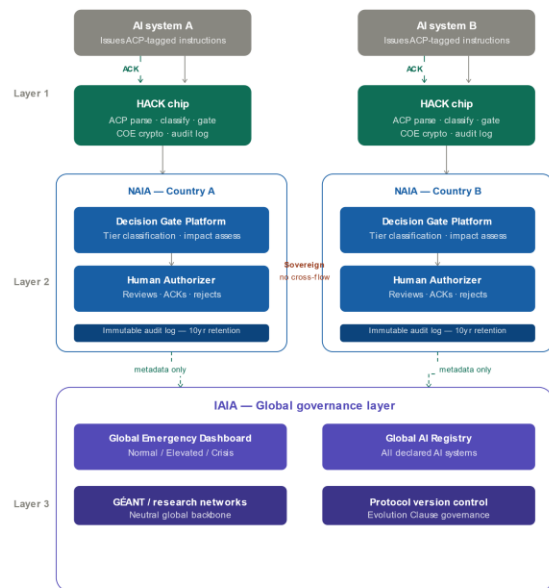
Endorsement key (EK)

Fused at manufacture — FIPS 140-3

RISC-V silicon root of trust

Three Layers: Silicon → National → Global

Enforceable without being centralized. Universal without requiring any state to surrender sovereignty.



National data flows upward — never sideways. Sovereignty by design.
Every AI action: silicon → national → global. Human authorized at Layer 2.

Figure 3 — Every AI action: silicon → national authority → global governance. Human authorized at Layer 2. Data flows upward only — never laterally between nations.

NAIA and IAIA — The Institutional Model

National AI Authority (NAIA)

- Ministerial-level Representative Lead
- Maintains national AI systems registry
- Audits Decision Gate compliance
- Manages AI incidents and enforcement
- Reports compliance metadata to IAIA
- National data remains under national jurisdiction at all times

International AI Agency (IAIA)

- UN specialized agency — modelled on IAEA
- Equal voting rights — no permanent-member veto
- Operates the Global AI Registry
- Global Emergency Dashboard (Normal/Elevated/Crisis)
- Coordinates with IAEA, WHO, BIS/FSB, ITU
- Funded by member states + chip micro-levy at manufacture

No single actor can override the IAIA alert level — it is set algorithmically. Industry contributions carry no voting rights or policy influence.

Sovereignty by Design

National data flows upward to IAIA — never laterally between nations. Each NAIA owns its data.

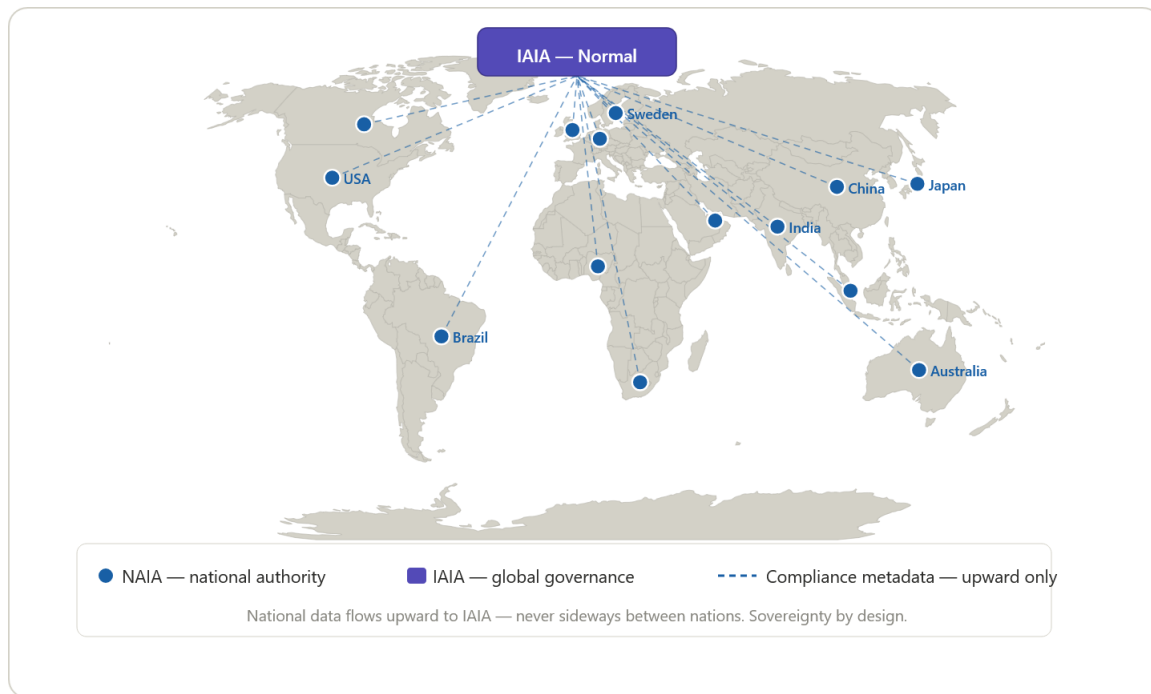


Figure 4 — Global IAIA/NAIA network. Compliance metadata flows upward from national authorities to the IAIA — never laterally between nations.

NAIA Operations Dashboard

• Judge Protocol — NAIA Operations Dashboard

• Normal Live · 14:37 UTC

HACK EVENTS TODAY

48,291

↑ 12% vs yesterday

PENDING AUTHORIZATIONS

7

Tier 2: 2 · Tier 3: 5

HARD STOPS TODAY

0

Last: 14 days ago

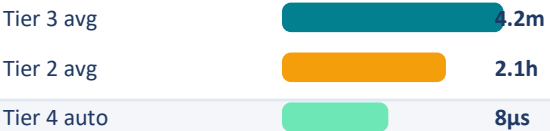
LIVE HACK EVENT STREAM

T4	ACK	Case classification — benefit eligibility	SE-TAX-01
T3	ACK	Administrative override — residency status	SE-MIG-04
T2	PEND	Multi-party authorization pending — asset freeze	SE-FIN-09
T3	REJ	Authorizer rejected — deportation order	SE-MIG-07

AUTHORIZATION OUTCOMES



ACK LATENCY BY TIER



Four Phases: Prove Before Mandate

1

Phase 1

Sandbox

0–12 months

Decision Gate Platform deployed in a founding national NAIA, one domain. Live audit trail, ACK latency, override rate data published as v1.0 evidence base.

2

Phase 2

Reference Implementation

12–24 months

Software stack packaged as open-source Judge Protocol Image, deployable on commodity hardware. IAIA interim secretariat established.

3

Phase 3

Hardened Deployment

24–48 months

TMO PCIe enforcement card deployed. Air-gapped appliance under CERN OHL. Full domain rule set coverage across all twelve domains.

4

Phase 4

Silicon Enforcement

48–72 months

HACK core embedded in AI accelerator silicon. Government procurement mandates. Universal silicon-level AI governance enforcement.

The sandbox generates the evidence base that justifies mandatory adoption. Phase 1 asks: one nation, one domain, twelve months.

The Field Has Arrived at the Gate

Dario Amodei, Anthropic (June 10, 2026)

"Policy on the AI Exponential" — mandatory third-party testing, government power to block unsafe releases, IAEA-style coalition, G7 agenda. Backed by \$350M.

Pope Leo XIV (May 2026)

Magnifica Humanitas — "not permissible" to entrust lethal AI decisions. Calls for robust legal frameworks. Rome Call principles now elevated to binding moral obligation.

28 governments (2023)

Bletchley Declaration — multilateral acknowledgement of frontier AI risk. No enforcement mechanism. The gap the Judge Protocol fills.

The Gap

Testing gates the release.

Authorization gates the action.

Every proposed regime lacks a runtime verification layer.

When one government can switch off global AI without control — June 2026, US export directive, Fable 5 — the case for international authorization infrastructure is no longer theoretical

The Judge Protocol — published March 2026 — specifies one.

Three Open Invitations

1

Founding Sandbox Partner

One national authority. One domain. Twelve months. Deploy the Decision Gate Platform, publish the evidence base, become the founding node of the global network. Sweden is the natural first mover.

2

CERN / Hardware Collaboration

The HACK specification requires a neutral technical authority. CERN played this role for the Web. HLD Sections 4–8 are open for co-development under CERN OHL-S v2 via Openlab.

3

Military Exception Working Group

The most sensitive open question. Must be resolved with ICRC, UN GGE on LAWS, and member state defense establishments before v1.0 ratification. Experts invited.

The Judge Protocol — Enterprise Architecture Framework



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AI may recommend

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↓ human must authorize